

Lecture#05

Conditional Probability

The probability of an event A when it is known that some other event B has occurred is called a conditional probability and is denoted by $P(A|B)$. The symbol $P(A|B)$ is usually read as ‘the probability that A occurs given that B occurs or simply probability of A given B, where the slash ‘|’ stands for ‘given that’. In general $P(A|B)$ is not equal to $P(A)$.

With two events A and B, the most fundamental formula to compute conditional probability for A given B is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

and that for B given A is

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) \neq 0$$

It thus follows from the above equations that for two dependent events A and B,

$$P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$$

since $A \cap B = B \cap A$

This rule is frequently referred to as the multiplication law, multiplication theorem or law of compound probability.

Definition: For two events A and B, the probability of their simultaneous occurrence is equal to the product of the unconditional probability of A and conditional probability of B, given that A has actually occurred. Symbolically

$$P(A \cap B) = P(A)P(B|A).$$

We can extend the above multiplication rule when the number of events is 3 or more. For 3 events A_1 , A_2 and A_3 ,

$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2)$$

Where $C|A \cap B$ is read as “C occurs given that A and B have already occurred”.

For k events, the rule is as follows:

$$P(A_1 \cap A_2 \cap \dots \cap A_K) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \dots P(A_K|A_1 \cap A_2 \dots \cap A_{K-1}).$$

Problem 01: A pair of dice is thrown. Find the probability that sum of the points on the two dice is 10 or greater if a 5 appears on the first die.

Solution: Let A be the event that sum of the points on the two dice is 10 or greater and B be the event that a 5 appears on the first toss. Symbolically, we want to evaluate the conditional probability $P(A|B)$.

Now,

$$A = \{(4,6), (5,5), (5,6), (6,4), (6,5), (6,6)\},$$

$$B = \{(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)\}$$

$$A \cap B = \{(5,5), (5,6)\}$$

Hence,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/36}{6/36} = \frac{1}{3}$$

Problem 02: The probability that a married man watches a certain TV show is 0.4 and that his wife watches the show is 0.5. The probability that a man watches the show, given that his wife does, is 0.7. Find

- (a) The probability that a married couple watches the show.
- (b) The probability that a wife watches the show given that her husband does.
- (c) The probability that at least one of the partners will watch the show.

Solution: let us define two events H and W as follows:

H: Husband watches the show

W: Wife watches the show

We are given that

$$P(H)=0.4, P(W)=0.5 \text{ and } P(H|W) = 0.7.$$

- (a) The probability that the couple watches the show is

$$P(W \cap H) = P(W)P(H|W) = 0.5 \times 0.7 = 0.35$$

(b) The conditional probability that a wife watches the show given that her husband also watches

$$P(W|H) = \frac{P(W \cap H)}{P(H)} = \frac{0.35}{0.40} = 0.875$$

(c) The probability that at least one (either H or W or both) watches

$$P(W \cup H) = P(W) + P(H) - P(W \cap H) = 0.40 + 0.50 - 0.35 = 0.55.$$

Problem 03: A coin is tossed until a head appears or it has been tossed three times. Given that the head does not appear on the first toss, what is the probability that the coin is tossed three times?

Solution: A sample space for the experiment is $S = \{H, TH, TTH, TTT\}$.

The associated probabilities are

$$P(H) = \frac{1}{2}, P(TH) = \frac{1}{4}, P(TTH) = \frac{1}{8}, P(TTT) = \frac{1}{8}.$$

Let A be the event that the coin is tossed 3 times and B be the event that no heads appear in the first toss so that

$$A = \{TTH, TTT\}, B = \{TH, TTH, TTT\},$$

and hence

$$A \cap B = \{TTH, TTT\}.$$

The associated probabilities are

$$P(A) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}, P(B) = \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$$

and thus $P(A \cap B) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$.

Hence the required conditional probability is

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1}{2}.$$

Problem 04: In a community there are equal number of males and females. Suppose 5% of the males and 2% of females are disabled. A person is chosen at random. If this person is male, what is the probability that he is disabled?

Solution: Let D stand for the event ‘disabled’ and M and F respectively for male and female. As males and females are in equal proportion,

$$P(M) = P(F) = 0.5 .$$

Also

$$P(M \cap D) = 0.05 \times 0.5 = 0.025, \quad P(F \cap D) = 0.02 \times 0.5 = 0.01$$

We want the conditional probability that the selected person is disabled:

$$P(D|M) = \frac{P(D \cap M)}{P(M)} = \frac{P(M \cap D)}{P(M)} = \frac{0.025}{0.5} = 0.05$$

Bayes’ Theorem

The Bayes’ Theorem is a mathematical formula used to determine the conditional probability of events. Let A and B be two events and let $P(A|B)$ be the conditional probability of A given B has occurred. Then, Bayes’ theorem states that-

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}$$

Proof: First, consider the following facts-

According to conditional probability:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$\Rightarrow P(A \cap B) = P(B|A)P(A) \rightarrow (1)$$

Now, using total probability theorem -

$$P(B) = P((B \cap A) \cup (B \cap A^c))$$

$$\Rightarrow P(B) = P(B \cap A) + P(B \cap A^c)$$

$$\Rightarrow P(B) = P(B|A)P(A) + P(B|A^c)P(A^c) \rightarrow (2)$$

Note that, $A \cap B = B \cap A$

Now,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(A)}; [Using \rightarrow (1)]$$

$$\Rightarrow P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}; [Using \rightarrow (2)]$$

Proved

For the events A, B, C, D ;

$$P(A|D) = \frac{P(A)P(D|A)}{P(D)} = \frac{P(A)P(D|A)}{P(A)P(D|A) + P(B)P(D|B) + P(C)P(D|C)}$$

In general, for k events $A_1, A_2, \dots, A_k$;

$$P(A|D) = \frac{P(A_i)P(D|A_i)}{\sum_{i=1}^k P(A_i)P(D|A_i)}$$

Problem 1: A blood test is 90% effective in detecting a certain disease when the disease is present. However, the test yields a false-positive result for 5% of the healthy patients tested. Suppose 1% of the population has the disease. Find the conditional probability that a randomly chosen person actually has the disease given that his test result is positive.

Solution: Let,

D: the person has the disease

P: the test is positive

Now,

$$P(D | P) = \frac{P(D \cap P)}{P(P)} = \frac{P(D)P(P | D)}{P(D)P(P | D) + P(D^c)P(P | D^c)}$$

$$\Rightarrow P(D | P) = \frac{0.01 \times 0.90}{(0.01 \times 0.90) + (0.99 \times 0.05)} = 0.15$$

Problem 2: Consider two containers. The first container contains 2 white and 7 black balls and second container contains 5 white & 6 black balls. We flip a fair coin and then draw a ball from the first container or the second container depending on whether the outcome was

heads or tails. What is the conditional probability that the outcomes of the toss were heads given that a white ball was selected?

Solution: Let,

W: the event that a white ball is drawn

H: the event that the coin comes up heads

Now,

$$\begin{aligned}P(H | W) &= \frac{P(H \cap W)}{P(W)} = \frac{P(H)P(W | H)}{P(W)} \\&= \frac{P(H)P(W | H)}{P(H)P(W | H) + P(H^c)P(W | H^c)} \\&= \frac{\left(\frac{1}{2}\right)\left(\frac{2}{9}\right)}{\left(\frac{1}{2}\right)\left(\frac{2}{9}\right) + \left(\frac{1}{2}\right)\left(\frac{5}{11}\right)} = \frac{22}{67}.\end{aligned}$$

Problem 03. A person uses his car 30% of the time, walks 30% of the time, and rides the bus 40% of the time to go to work. He is late 10% of the time when he walks, 3% of the time when he drives, and 7% of the time when he takes the bus.

- (i) What is the probability that he will be late?
- (ii) If he is late, what is the probability that he took the bus?
- (iii) If he is late, what is the probability that he took the car?

Problem 03. A health program sends patients to one of three testing centers. Center A tests 50% of patients, Center B tests 20% of patients, and Center C tests 30% of patients. For patients who actually have the disease, the probability the center's test comes back positive is: Center A: 2% detection, Center B: 12% detection, and Center C: 6% detection.

- (i) What is the probability that the test is positive?
- (ii) If a patient's test is positive, what is the probability that the test was tested at Center B?
- (iii) If a patient's test is positive, what is the probability that the test was tested at Center B?

Problem 04. A clinic uses one of three screening methods for a condition. The blood test is used for 40% of patients. The imaging scan is used for 35% of patients. The physical exam

is used for 25% of patients. When a patient actually has the condition, the methods detect it with these probabilities: Blood test detects it in 80% of cases, Imaging detects it in 70% of cases, and Physical exam detects it in 90% of cases.

- (i) What is the probability that a randomly chosen patient who has the condition will be detected (i.e., screened positive)?
- (ii) If a patient who has the condition is detected, what is the probability the method used was the physical exam?
- (iii) If a patient who has the condition is detected, what is the probability the method used was the blood test?

Problem 05. A student prepares for exams using three methods. He studies from books 40% of the time, he watches video lectures 35% of the time, and he attends group study sessions 25% of the time. He passes a quiz 80% of the time when he studies from books, 70% of the time when he watches video lectures, and 90% of the time when he attends group study.

- (i) What is the probability that he passes the quiz?
- (ii) If he passed, what is the probability that he used group study?
- (iii) If he passed, what is the probability that he used books?